

OmniMed-FL: A Robust Multimodal Federated Learning Framework for Clinical Diagnosis

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Abstract—We present OmniMed-FL, a controlled multimodal federated-learning systems study for five-class image–text classification. The proxy corpus contains 2,400 public chest radiographs, 600 synthetic COVID-19 image surrogates, and 3,000 class-conditioned synthetic notes paired to images by class rather than patient. Public DistilBERT and ViT-Base/16 encoders are fine-tuned with random task modules; no pooled-client model initializes the operational runs. We vary Dirichlet skew, client count, anti-collapse components, initialization, eight operational fusion rules, missing text, and matched federated controls, while reporting runtime, peak allocated GPU memory, and formula-derived FP32 traffic. At $K=5$ and severe skew ($\alpha=0.1$), local-only training obtains 0.310 Macro F1 (five-client mean), versus 0.652 ± 0.057 for FedAvg, 0.685 ± 0.064 for FedProx, and 0.192 ± 0.129 for our SCAFFOLD-style AdamW adaptation; the 0.033 FedProx–FedAvg gap is smaller than either two-seed sample standard deviation. In the heterogeneity sweep, FedAvg changes from 0.896 ± 0.004 at $\alpha=5$ to 0.674 ± 0.185 at $\alpha=0.1$. A branch audit records 0.248/0.324/0.573 GiB FP32 payload per client per round for text/image/multimodal models. Because the notes and one image class are synthetic, pairing is not patient-level, and no external clinical validation is performed, these are descriptive proxy comparisons rather than estimates of diagnostic performance, privacy, or deployment readiness.

Index Terms—Multimodal Federated Learning, Non-IID Data, Multimodal Fusion, Clinical Classification, Communication Cost, Retrieval

I. INTRODUCTION

CLINICAL prediction often draws on both images and textual context. Centralized image models [1], [2] and medical vision–language models can combine these signals, but cross-institutional training would ordinarily require data to leave its source. Federated learning (FL) instead exchanges parameter updates while retaining each client’s raw examples [3]. This data-locality property is useful, although FL alone is not a privacy guarantee and does not remove risks from model updates.

Federated medical imaging [4]–[6] and centralized medical vision–language modeling have established precedents, with recent work also studying their intersection. The combination still raises questions that a single favorable run cannot answer: how performance changes with client skew, whether anti-collapse components help, whether fusion rankings survive seed

variation, and what additional memory and communication the multimodal branch incurs.

We study these questions through OmniMed-FL, a controlled five-class proxy benchmark. Its public radiographs and synthetic clinical-style notes are not patient-matched data, so the work evaluates the learning system rather than clinical diagnosis. This scope permits internally matched comparisons while making the boundary between a systems result and a clinical claim explicit. Here, “robust” denotes stress testing across partitions, client counts, and missing modalities, not certified clinical robustness.

A. Contribution

The main *contributions* of this paper are as follows:

- We implement a controlled multimodal FL protocol that begins from public pretrained encoders and random task modules, never from a model trained on pooled client examples.
- We evaluate heterogeneity, client count, anti-collapse components, initialization, and eight fusion rules under one multi-round FL protocol. Separate matched adaptations cover FedMME’s one-shot and P-FIN’s missing-text settings.
- We report realized client skew, wall-clock time, peak allocated GPU memory and FP32 communication payload, and evaluate retrieval over the full held-out split rather than a five-query demonstration.

II. RELATED WORK

Centralized radiograph classifiers [1], [2] and Vision Transformers [7], [8] provide established image-only precedents. The clinical FL studies cited here are also image-only: Sheller *et al.* evaluate federated averaging, Dou *et al.* conduct multinational external validation, and Kaissis *et al.* combine differential privacy, secure aggregation, and encrypted inference [4]–[6]. FedProx and SCAFFOLD address heterogeneous-client optimization through proximal regularization and control variates, respectively [9], [10].

BiomedCLIP and MedCLIP learn medical image–text representations, while LLaVA-Med instruction-tunes a biomedical

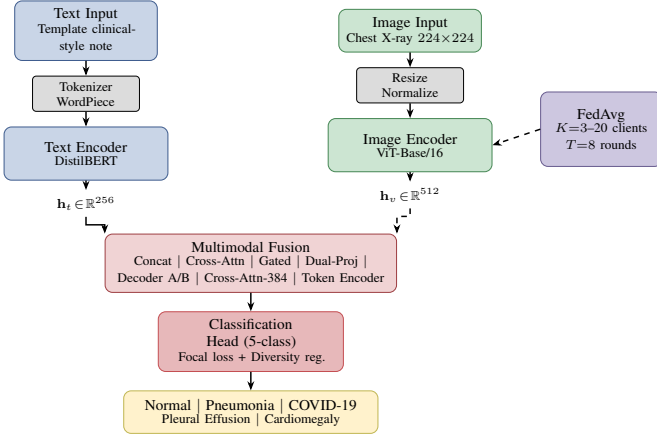


Fig. 1. OmniMed-FL architecture. DistilBERT and ViT-Base/16 encode the two proxy modalities; lightweight multimodal fusion combines them; FedAvg coordinates simulated clients without exchanging their raw examples.

vision–language assistant; all three are centralized [11]–[13]. Recent multimodal FL includes FedMME’s one-shot hard-voting ensemble and P-FIN’s uncertainty-aware feature imputation for missing modalities [14], [15]. Because their published datasets, backbones, tasks, and budgets differ, we report clearly labeled adaptations on our common corpus and local-epoch budget rather than compare cross-paper scores or claim an exact reproduction.

III. SYSTEM DESIGN

A. Architecture

Figure 1 shows the OmniMed-FL pipeline. Clinical-style notes are tokenized with WordPiece to 128 tokens; chest images are resized to 224×224 and ImageNet-normalized. DistilBERT’s [16] [CLS] representation is projected to $\mathbf{h}_t \in \mathbb{R}^{256}$, while the mean-pooled ViT token sequence is projected to $\mathbf{h}_v \in \mathbb{R}^{512}$.

The two representations are combined by eight lightweight operators: projected concatenation, residual cross-attention, gated residual fusion, dual-projection concatenation, one-layer decoder A, a distinct two-query/two-layer decoder B, non-residual 384-dimensional cross-attention, and a two-token Transformer encoder. Historical code keys resemble foundation-model names, but these are compact fusion rules rather than implementations of those models. Each feeds the same two-layer five-class MLP, and all eight are compared inside the operational FL loop with the same data, partition, optimizer, and local budget.

B. Federated and Local Objectives

Let K clients hold disjoint shards $\mathcal{D}_k$ of the training split, with $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$, and let θ collect the trainable parameters on the fusion path: both encoders, the projection

layers, the fusion operator and the classification head. The federated objective is the sample-weighted empirical risk

$$\min_{\theta} F(\theta) = \sum_{k=1}^K \frac{n_k}{n} F_k(\theta), \quad (1)$$

$$F_k(\theta) = \frac{1}{n_k} \sum_{(x,z,y) \in \mathcal{D}_k} \ell(f_{\theta}(x,z), y).$$

where x is a radiograph, z its class-paired note and f_{θ} the fused five-class classifier. Shards are never exchanged; the server receives only the trainable tensors enumerated above.

The local loss combines a focal term with a batch-level diversity penalty,

$$\ell = (1 - p_y)^{\gamma} \text{CE}_{\epsilon}(f_{\theta}(x,z), y) + \lambda \left(1 - \frac{H(\bar{p})}{\log C}\right), \quad (2)$$

with focusing parameter $\gamma=2$, label smoothing $\epsilon=0.05$, $C=5$ classes, and $\bar{p}$ the mean predicted distribution over the mini-batch. The second term grows when a client’s predictions concentrate on few classes, a behavior that can occur under severe Dirichlet skew. Setting $\lambda=0$ removes it; together with a class-balanced batch sampler it forms the anti-collapse stack ablated in Section IV.

At each round, the server broadcasts $\theta^{(t)}$; each eligible client performs three class-balanced local AdamW epochs using (2), clips the gradient norm at one, and returns its trainable tensors. The server applies the sample-weighted update in (1). Operational runs begin with public DistilBERT and ViT-Base/16 encoder weights plus random projection, fusion, and classification heads; no pooled-client model initializes them. Both encoders are fine-tuned, whereas unused unimodal heads and the inactive CNN fallback are excluded from optimization and communication.

C. Controlled Proxy Corpus

Image data. We form a balanced 3,000-image proxy corpus (600 examples per class). The public radiographs come from the Kermany pneumonia collection [17] and the NIH ChestX-ray dataset [18]. Loader logs verify 2,400 public images covering Normal, Pneumonia, Pleural Effusion and Cardiomegaly. The originally specified COVID-19 source was unavailable at execution time, so the 600 COVID-19 images are synthetic X-ray-style surrogates with bilateral opacity patterns. Images are resized to 224×224 , normalized with ImageNet statistics, shuffled deterministically and split 80/20 into 2,400 training and 600 validation examples.

Clinical text data. No patient EHR data are used. Each image label is paired at the class level with a synthetic clinical-style note generated from observation, symptom, context and indicator slots. Twenty-five percent of notes draw all slots from the target class, 35% mix target and non-target slots, and 40% are heavily mixed while retaining a probabilistic target-class indicator. This controlled construction reduces direct label wording, but the image and note are not from the same patient and must not be described as a real image–report pair.

What this corpus does and does not support. Every experimental arm sees the same controlled corpus, which supports

internally controlled, descriptive comparisons between federated algorithms, fusion rules and ablations. The absolute scores are not diagnostic-performance estimates: class-conditioned templates can create shortcuts, label-level pairing removes real patient discordance and missingness, and the synthetic COVID-19 surrogates may not resemble clinical COVID-19 radiographs. The split is also not patient-grouped across source institutions. We therefore make no claim about clinical utility, deployment readiness, or generalization to real paired reports.

D. Experimental Setup

Hyperparameters. Each client uses AdamW with learning rate 10^{-4} and weight decay 0.01 for three local epochs per round. The reviewer suite uses eight rounds, batch size 16, FP32 tensors without automatic mixed precision, and diversity-loss weight 1.0. The reference setting is five clients with Dirichlet $\alpha = 1.0$; Section IV varies α and the number of clients explicitly. Seeds 0 and 1 jointly control task-head initialization and the realized client partition. Experiments run in PyTorch on one NVIDIA H100 NVL device in a server shared with unrelated workloads. The GPU-synchronized timer covers sequential local training and aggregation within each round; validation, setup, I/O, and networking are excluded, while server contention remains uncontrolled. CUDA deterministic algorithms are not forced, so seeds control software RNG streams without guaranteeing bitwise-identical reruns. Simulated clients train sequentially on one GPU, so time is an execution-cost record rather than distributed throughput. Table I records the fixed protocol.

Partitioning. For each class, a Dirichlet draw allocates training indices across clients. Smaller α creates more concentrated client label distributions; $\alpha \rightarrow \infty$ approaches IID allocation [19]. Because nominal α does not fully describe a finite realized split, we also report the mean client share occupied by its dominant class. The same split is held fixed across rounds for a given seed. Shards with fewer than four training examples are not locally updated; scalability results therefore report realized active and nominal K .

Matched recent-method controls. Our matched FedMME-style arm uses 24 local epochs rather than FedMME’s native 100, uploads each client model once, and applies its equal-weight hard voting; our implementation breaks ties by mean softmax [14]. This preserves the common corpus, shards, encoders, optimizer, and matched local compute, but substitutes our five-class inputs for FedMME’s generated-report pipeline. The P-FIN-style stress test fixes three multimodal and two image-only clients, projects both encoders to 256 normalized features, and compares zero fill, deterministic FIN, Gaussian beta-NLL imputation ($\beta=0.5$), and uncertainty-weighted aggregation with Fed-UQ-Avg balance weight 0.6 and temperature 0.2 [15]. It uses the common concat classifier rather than P-FIN’s native fusion stack. FedProx uses $\mu=0.01$. These are matched adaptations, not exact reproductions; their native scopes and communication schedules remain different.

Rationale for Table I. Within each ablation or comparison family, only the named factor changes. Three local epochs and

TABLE I
COMMON SIMULATION PARAMETERS.

Parameter	Value
Local Epochs (E)	3
Federated Rounds (T)	8
Batch Size	16
AdamW Learning Rate	1×10^{-4}
Weight Decay	0.01
Diversity Reg. (λ)	1.0
Systems-grid K (reference)	3, 5, 10, 20 (5)
Systems-grid α (reference)	0.1, 1, 5 (1)
$K=5$ additional α	0.3, 0.5
Seeds	0, 1
Initialization	Public encoders + random task modules
Precision	FP32 tensors (no AMP)

eight rounds give a tractable 24-local-epoch budget, while batch size 16 holds both encoders and one local model copy in GPU memory. The Cartesian systems grid crosses $K=\{3, 5, 10, 20\}$ with $\alpha=\{0.1, 1, 5\}$; the $K=5$ heterogeneity sweep additionally includes $\alpha=0.3$ and 0.5. Thus the study spans severe to near-IID label skew, with $K=5$, $\alpha=1$ as its reference. Two seeds were computationally feasible and support descriptive, not significance, claims. The 10^{-4} rate and $\lambda=1$ are fixed implemented defaults, not per-model optima.

IV. RESULTS

A. Matched Performance and Ablations

At $\alpha=0.1$, local-only averages 0.310 Macro F1 across five clients in one partition. Over two matched seeds, FedAvg obtains 0.652 ± 0.057 and FedProx 0.685 ± 0.064 ; their 0.033 gap is smaller than either sample SD (Fig. 2(a)). SCAFFOLD-AdamW gives 0.192 ± 0.129 and oscillates in panel (b), indicating instability of our AdamW control-variate adaptation, not classical SGD SCAFFOLD. The FedMME-style arm trains five independent client models for 24 epochs and combines predictions by equal hard plurality with mean-softmax tie breaking; it preserves the one-shot idea but is not an exact reproduction.

Under severe skew, the full balanced-loader/diversity-loss stack reaches 0.716 ± 0.014 F1; removing the loader, diversity term, or both gives 0.585 ± 0.148 , 0.701 ± 0.011 , and 0.657 ± 0.057 , respectively (panel (c)). Diversity diamonds separate transient class loss from lower F1; at $\alpha=1$, no directional component effect is resolved with two seeds. Independent baseline, heterogeneity-sweep, and ablation executions of the nominally repeated severe-skew FedAvg/concat setting yield 0.652 ± 0.057 , 0.674 ± 0.185 , and 0.716 ± 0.014 ; none is pooled or selected. Panel (d) instead trains all eight fusion rules in the operational FL loop.

All-random retains the DistilBERT/ViT architectures but randomizes encoder and task weights. All three initialization arms then use the same 8×3 FL schedule; pooled-start alone reuses one seed-0 checkpoint selected by a 12-epoch centralized pooled run for both FL seeds, so its extra pretraining is outside that budget. It starts at 0.843 rather than 0.704 round-1 F1 but ends at 0.818 rather than 0.839; only this labeled ablation uses the pooled checkpoint. The separated abort-off/on controls

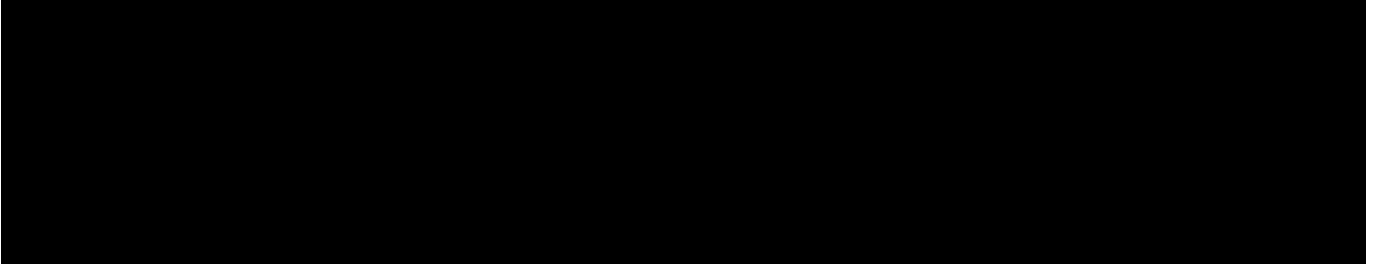


Fig. 2. Matched results (two-seed mean $\pm$ sample SD unless noted). (a) Severe-skew baselines: local-only is the five-client mean from one partition; FedMME-style is a one-shot hard-voting ensemble after 24 independent local epochs, whereas iterative arms use 8 rounds of 3 epochs. (b) Iterative convergence. (c) Anti-collapse F1 with diamonds for minimum predicted-class diversity. (d) Eight operational fusion rules. (e) Federated initialization (round-1 diamonds) and a separated, non-federated 12-epoch pooled abort control. (f) P-FIN-style adaptation with three multimodal and two image-only clients; all held-out text is forced missing for the primary metric, while full-text F1 is secondary.

(0.845 ± 0.013 and 0.884 ± 0.003) are non-federated runs using the unbalanced pooled loader with diversity loss disabled. Every run completes 12 epochs with minimum diversity 1.0, so the LR-boost/abort branch never fires and the difference is nondeterministic rerun variation, not an early-abort effect. Panel (f) is a matched P-FIN-style adaptation: 256-dimensional L_2 -normalized features and concat+MLP replace the source bidirectional attention; forced-missing text is primary and full-text performance secondary.

B. Scalability, Communication, and Resource Cost

Figure 3(a) crosses $K=3, 5, 10, 20$ with $\alpha=0.1, 1, 5$. At $K=5$, F1 falls from 0.896 ± 0.004 near IID ($\alpha=5$) to 0.674 ± 0.185 under severe skew; auxiliary $\alpha=0.3, 0.5$ checks give 0.769 ± 0.043 and 0.824 ± 0.113 . This spread and the realized allocation in panel (f) show why nominal α and one seed are insufficient. Cells report A/K (or the two-seed A range); shards smaller than four samples skip local optimization.

Panels (b)–(c) time the GPU-synchronized sequential local-training/aggregation block and its peak allocated memory on one shared H100 NVL; validation, setup, I/O, networking, concurrency, and multi-node throughput are excluded. For $|\theta|$ trainable parameters, b bytes/parameter, T rounds, and nominal K , the plotted bidirectional volume is

$$V_{\text{nom}} = 2KT|\theta|b, \quad (3)$$

Here $|\theta|=153,935,621$, $b=4$, and $T=8$, so the 615,742,484-byte payload gives 27.5/45.9/91.8/183.5 GiB for $K=3/5/10/20$ (1 GiB = 2^{30} bytes). This nominal count remains when $A < K$; replacing K by A is an alternative estimate, not a measurement. Serialization, headers, retransmission, secure aggregation, compression, and algorithm-specific state are excluded.

In the one-run branch audit (panel (e)), text/image/multimodal F1 is 0.893/0.740/0.782; FP32 payload is 0.248/0.324/0.573 GiB per client/round, timed-round sums are 18.5/33.6/45.0 min, and peak allocated memory is 1.69/3.18/4.77 GiB. The text branch exceeding concatenation means this proxy run does not support blanket multimodal superiority; class-conditioned template shortcuts are one plausible cause.

C. Retrieval Evidence for the RAG Stage

Exact inner-product FAISS search [20] over TF-IDF note vectors gives condition-macro averages of 0.522 top-1 same-label accuracy, 0.462 same-label precision@5, and 0.707 mean top-1 cosine similarity across the five condition rows of the 600-query evaluation (Fig. 4). This replaces the earlier five-query illustration. It is a synthetic-note retrieval check, not a fine-tuned-DistilBERT result, generated-answer evaluation, or evidence of clinical retrieval quality.

V. DISCUSSION AND LIMITATIONS

The experiments support systems-level conclusions only. Strong partition sensitivity persists despite anti-collapse measures: at $\alpha=0.1$, even identically configured FedAvg reruns vary substantially, and the audited runner skips local optimization for shards with fewer than four samples. The client-count study is a sequential simulation; it does not model concurrent workers, network latency, stragglers, intermittent dropout, or heterogeneous accelerators. Robust aggregation against poisoned or Byzantine updates is also untested. Accordingly, “robust” in the title denotes the reported stress tests, not a formal guarantee.

Communication is a second boundary. Equation (3) exposes model-state scaling, while Fig. 3(e) shows the larger multimodal payload and memory footprint. A deployment study must additionally measure end-to-end traffic and latency under compression, partial participation, failures, and secure aggregation. This simulation implements neither secure aggregation nor the differentially private training and privacy accounting of Abadi *et al.* [21]. It also does not evaluate attacks on model updates; retaining raw examples locally is therefore not a formal privacy guarantee.

Clinical interpretation is more restricted. All 3,000 notes and the 600-image COVID-19 class are synthetic; the other 2,400 images are public radiographs. Image-text pairing is by class rather than patient; class-conditioned templates can encode label shortcuts; and label-level pairing removes the discordance, missingness, temporal context, and site effects of real records. The split is not patient- or institution-grouped, and there is no external or prospective validation. Thus neither absolute F1

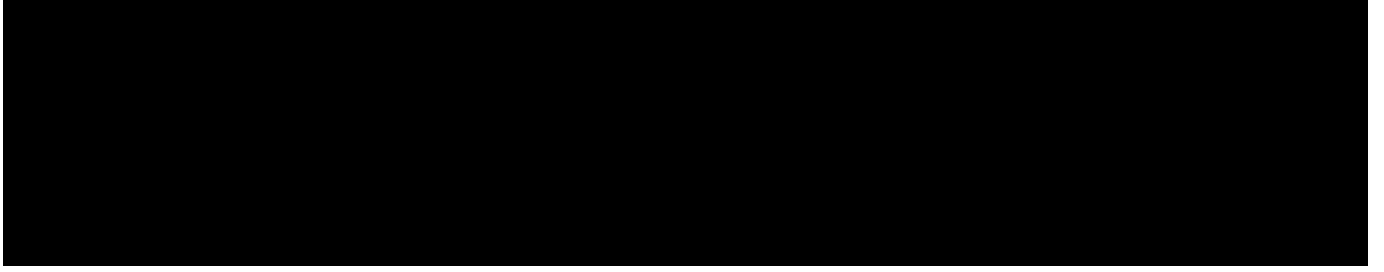


Fig. 3. Scalability/resource audit. Two-seed mean $\pm$ sample SD is shown for (a) Macro F1 (A/K is active/nominal clients or the two-seed A range), (b) the GPU-synchronized training/aggregation section per round, and (c) peak allocated memory within it. (d) Formula FP32 volume counts one download and upload for every nominal client in eight rounds, including locally skipped shards; it is not measured traffic. (e) One-run branch cost at $\alpha=1$, $K=5$ (timed-round sum and per-client/round payload). (f) Severe-skew seed-0 counts, colored by within-client class share.

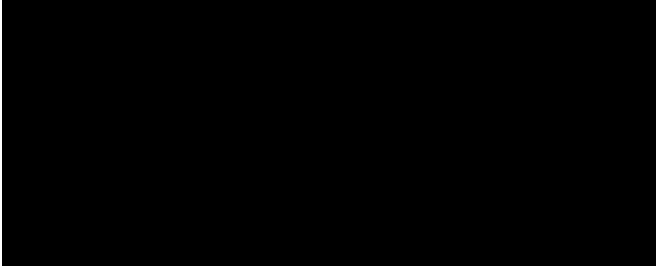


Fig. 4. Exact TF-IDF/FAISS retrieval over all 600 held-out queries. Bars show top-1 same-condition-label accuracy and same-condition-label precision@5; diamonds show mean top-1 cosine similarity with standard-deviation whiskers. This evaluates retrieval, not generation.

nor retrieval scores estimate diagnostic safety, generalization, clinical utility, or deployment readiness.

Finally, two seeded runs without enforced deterministic CUDA algorithms permit descriptive means and sample standard deviations, not significance claims. The matched FedMME- and P-FIN-style arms retain important method ideas under a common corpus and 24-local-epoch budget, but use different architectures/schedules and substitute our five-class inputs and fusion stack; they are not exact reproductions of the source papers. Larger multi-site datasets, more seeds, native-method replications, calibrated uncertainty, privacy/security tests, and prospective evaluation are necessary next steps.

VI. CONCLUSION

OmniMed-FL is a controlled multimodal-FL systems study rather than a clinical validation. Reported performance and resource cost are conditional on partition realization and protocol choices; the fusion, initialization, and missing-modality comparisons are descriptive stress tests. Real patient-paired multi-institutional studies and deployment safeguards remain necessary before clinical use.

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